

Hypothesis Testing Overview

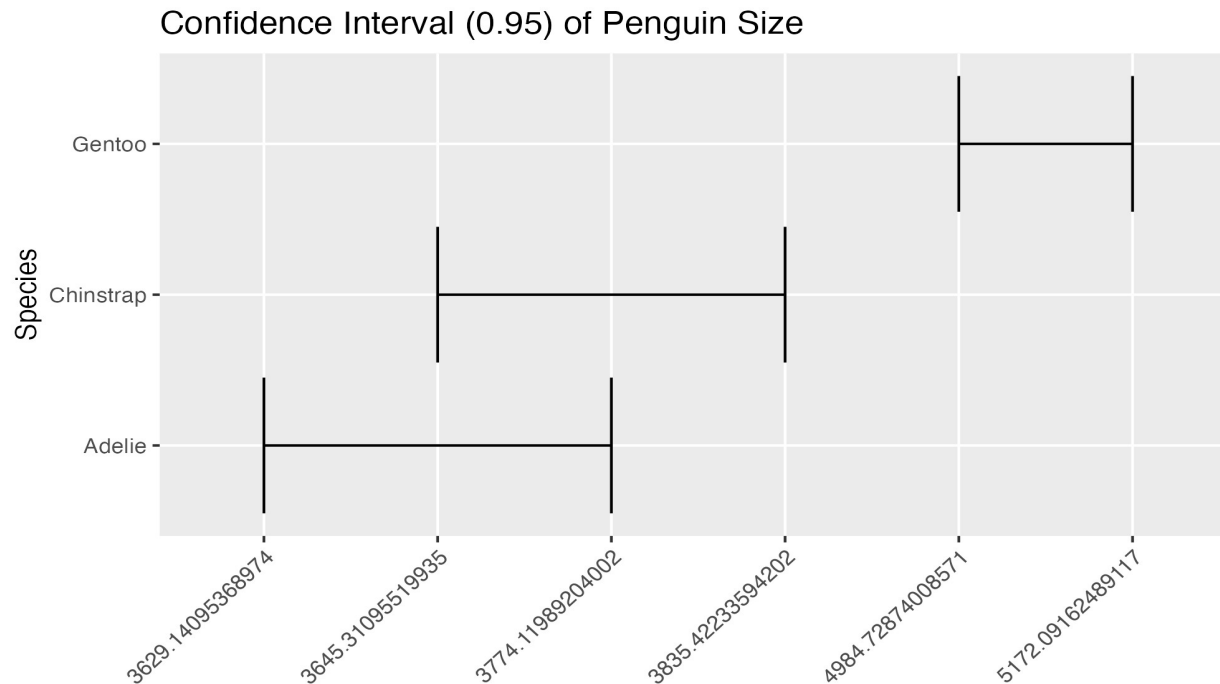
Exercise



The plot below shows the 95% confidence intervals for mean body mass of different penguin species.

Do you agree with the following statement:

Based on this plot alone, we can conclude that Chinstrap penguins have higher body mass on average than Adelie penguins.



- a) Strongly Agree
- b) Agree
- c) Neutral
- d) Disagree
- e) Strongly Disagree

Hypothesis Testing

A systematic framework for evaluating whether the evidence in our data is strong enough to reject a specified null hypothesis.

We start with an assumption to evaluate.

Starting assumption → **Null hypothesis**

Alternative explanation → **Alternative hypothesis**

Review the evidence → **Calculate a test statistic**

Is the evidence sufficient? → **Compare p-value significance level**

Draw a conclusion → **Reject or fail to reject H_0**

If the evidence is sufficient, we reject our starting assumption in favor of the alternative explanationn.

p-value

The probability of observing a result as extreme as or more extreme than ours, assuming H_0 is true.

Small p-value → evidence against H_0

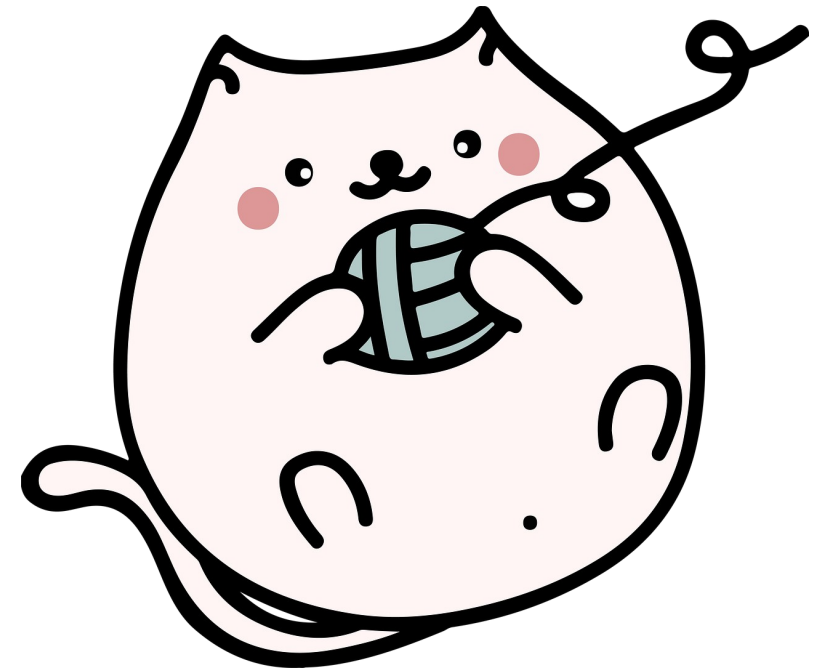
We have a preconceived notion about the world. All cats are meanies.

We are presented with an alternative view. Not all cats are meanies (note: don't ask my dog).

We review the evidence. I travel the entire universe and have a meet and greet with every single cat. About 78 percent of our encounters were very pleasant.

We determine if the evidence is enough. Are these meet and greets sufficient for capturing the overall personality of cats?

If the evidence is sufficient, we reject our preconceived notion and accept the alternative view as our new reality. I have sufficient evidence to reject the claim that all cats are meanies.



Hypothesis testing leads to one of two decisions: *Reject H_0 or Fail to reject H_0*



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